Year 8 Investigation 3

Linear Models

Baldivis Secondary College

Name: Solutions.

/ 40 marks
Time: 50 minutes

The school is planning a disco for all the year 9 students. It is your job to organize the music for the night. You have found the following 3 alternatives for the music.

Choice 1 - Rick's Rock Group
Rick charges $100 per hour and but charges a flat fee of $20 for coming out

Choice 2 - Magic Mark DJ service
Mark charges $40 per hour plus a $200 flat fee.

Choice 3 - Steve's Death Metal Band.
Steve charges $60 per hour plus a $100 flat fee.

1. Complete the table of values for each option and write a rule using "C" for cost and "N" for Number of hours that could be used for each option. **[8 marks]**

Rick's Rock Group

N (number of hours)	0	1	2	3	4	5	6
C (cost in $)	20	120	220	320	420	520	620

✓

Now write a rule you can use to calculate the cost of Rick's Rock Group.

$C =$ 100 n + 20 ✓

Magic Mark's DJ Service.

N (number of hours)	0	1	2	3	4	5	6
C (cost in $)	200	240	280	320	360	400	440

✓✓

Now write a rule you can use to calculate the cost Magic Mark's DJ Service.

$C =$ 40 n + 200 ✓

Steve's Death Metal Band.

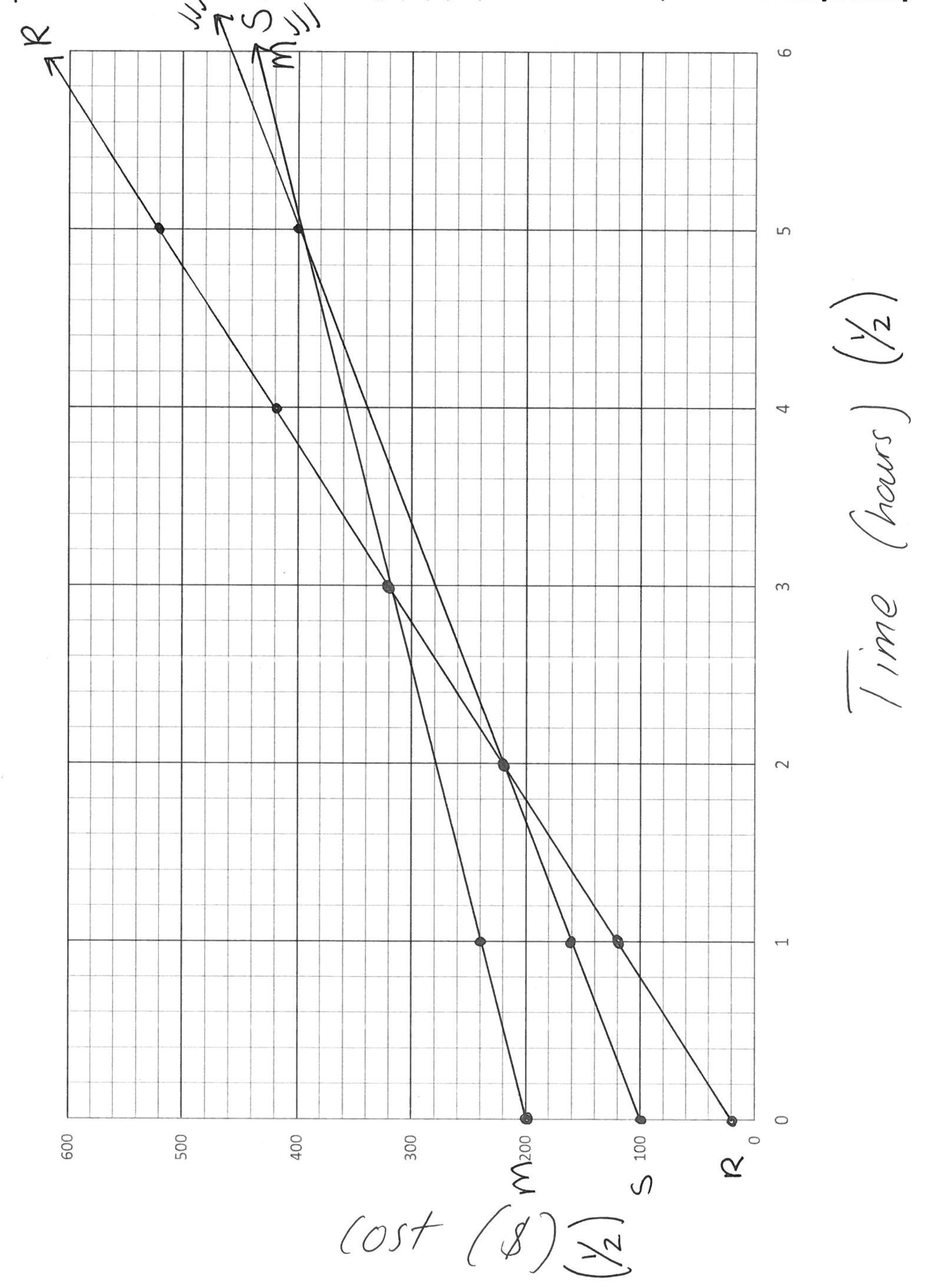
N (number of hours)	0	1	2	3	4	5	6
C (cost in $)	100	160	220	280	340	400	460

✓✓

Now write a rule you can use to calculate the cost Steve's Death Metal Band.

$C =$ 60n + 100 ✓

2. Plot these **three options** on the graph paper provided. **Label each option** and the **axes**. **[10 marks]**



3. Which music group is the cheapest in the first 2 hours? **[1 mark]**

Rick

4. Which music group is the cheapest after 4 hours? **[1 mark]**

Steve

5. Which music group is the cheapest after 6 hours? **[1 mark]**

Mark

6. Which option is the best value if the disco is to go for 1 hours? **[1 mark]**

Rick

7. How much will the cheapest music act cost if the disco goes for 4 hours? **[1 mark]**

$340

8.

a. The principal has set the times for the disco from 7pm until 10:30pm. **Using your graph**, find out what the cost would then be for each of the three options. **[3 marks]**

Rick's Rock Group	Magic Mark's DJ	Steve's Death Metal
$370	$340	$310

b. Check that these values are correct by substituting N (number of hours) back into your rule to find the cost. Show your working. **[6 marks]**

Rick's Rock Group

C = 100 × 3·5 + 20

C = 350 + 20

C = $ 370

✓✓

Magic Mark's DJ

C = ~~100~~ 40 × 3·5 + ~~20~~ 200

C = 140 + 200

C = $340

✓✓

Steve's Death Metal

C = ~~100~~ 60 × 3·5 + ~~20~~ 100

C = 210 + 100

C = $310

✓✓

9. If you are given a budget of $300, how many hours could you hire Magic ~~Mike~~ Mark's DJ service for? **[1 mark]**

2 ½ hrs

10. Sales of tickets has resulted in 30 tickets getting bought at $10 each. Using the total money raised which band would be able to play the longest? Explain your answer. **[2 marks]**

$\$10 \times 30 = \$300$ raised ✓

Steve play the longest ✓

11. If an additional 15 tickets are sold would this change which group you would choose? Explain your answer. **[2 mark]**

$\$10 \times 45 = \$450$ raised ✓

Changes to Mark ✓

12. If I were to pay Steve's Death Metal Band $1000 to play at my birthday how many hours could I hire them for? Create and solve a linear equation to prove your answer. **[3 marks]**

$$60n + 100 = 1000$$ ✓

$$60n = 900$$
$$\div 60 \quad \div 60$$ ✓

$$n = 15$$ ✓

You could hire them for 15hrs.

End of Assessment